

happiness and curiosity; e.g., Ashby, Isen, & Turken, 1999). On the basis of the optimal balance model and the iterative reprocessing model, these changes should interact to foster improvements in self-regulation. Research with adults indicates that mindfulness training does indeed improve performance on a variety of measures of self-regulation (e.g., Baer, 2003; Chambers, Lo, & Allen, 2008; Heeren, Van Broeck, & Philippot, 2009; Mind and Life Education Research Network, 2012; Zeidan, Johnson, Diamond, David, & Goolkasian, 2010; Zylowska et al., 2008) and that it is associated with greater activation in the PFC networks underlying self-regulation (see Hořizel et al., 2011, for a review). For example, in a randomized design, 7 weeks of mindfulness training lessened the tendency for negative stimuli to interfere with a simple cognitive task (with corresponding reductions in skin conductance responses to those stimuli), as compared to an active control group trained in relaxation meditation (Ortner, Kilner, & Zelazo, 2007). Following an 8-week MBSR course, participants showed increased activation in right dorsolateral and ventrolateral PFC when attending to present experiences (Farb et al., 2007). In addition, there is evidence for greater PFC activation during other tasks in individuals who are higher in mindfulness (Cresswell, Way, Eisenberger, & Lieberman, 2007) and in individuals who have more experience with meditation (Hořizel et al., 2007).

ADAPTING MINDFULNESS PRACTICES FOR CHILDREN AND ADOLESCENTS

Standard adult exercises have been adapted for use with children and adolescents to create developmentally appropriate ways to train the core aspects of mindfulness, including nonjudgmentally noticing one's moment-to-moment experiences, monitoring attention and redirecting attention when it has wandered, and nonreactively observing one's thoughts and feelings. A recent surge of research has revealed that children and adolescents readily engage in these activities and enjoy doing so (Biegel, Brown, Shapiro, & Schubert, 2009; Broderick & Metz, 2009; Huppert & Johnson, 2010; Schonert-Reichl & Lawlor, 2010; see Burke, 2010, for a review). As with adults, mindfulness training for children and adolescents typically occurs in small-group sessions that include a variety of activities, such as body scans, breathing exercises, and sitting meditations. To compensate for children's limited self-regulation skills, the lessons and individual training activities are shorter for younger participants: Adults may be able to attend to their breathing for 45 min, but 5-year-olds may only manage for 3 min (Burke, 2010). Mindfulness practices may also involve more movement-based activities like yoga stretches or rocking of one's body, because attempting to remain still for long periods may be so difficult for young children that it interferes with mindful reflection (e.g., Kaiser-Greenland, 2010). Instructions are typically simplified, and teachers may use props or concrete metaphors to help children understand the goals of mindfulness exercises. For example, to help children understand the notion of a body scan, a teacher may tell children that he or she will use a hula-hoop as a scanner, just like the scanner at the grocery store that shines a light on each item it passes. Once children have practiced with the hula-hoop, the teacher may lead a group exercise where children use an imaginary hula-hoop to scan their bodies (e.g., Johnson et al., 2011). Props may also help children focus on their breathing. For example, a teacher may place a stuffed animal on a child's abdomen and instruct the child to rock it to sleep with gentle breaths (Kaiser-Greenland, 2010). In addition, to foster children's mindful awareness of the sensation of touching, children may be asked to manipulate a common object that they are holding behind their back, and notice how the object feels (e.g., "What is the texture of the object? Is it smooth or rough?"). Such exercises provide an opportunity to emphasize easily overlooked aspects of their experience: If children touch the object and tell the teacher what the object is ("It's a key!"), the teacher may redirect their attention (saying, "I don't want you to tell me what it is—I just want you to notice how it feels to you"). Asking children to focus their attention on their sensations may lay the foundation for mindful awareness of more complex aspects of their subjective experience, such as emotions or thoughts. For example, children may be told that thoughts pass through the mind like floats pass by in a parade; some of the floats (thoughts) may grab their attention more than others, but just as they would not jump onto a float at a parade, they can simply observe their thoughts as they occur (Saltzman & Goldin, 2008). Throughout all of these activities, teachers' own behavior can serve as an important model for children's burgeoning mindfulness skills. In addition, parents, teachers, and caregivers may turn many daily experiences into opportunities to promote mindful awareness by prodding children to notice what is happening in the current moment in a purposeful and nonreactive way. For example, during meals, parents might challenge their children to reflect on the food they are eating: "Is it hot or cold? Is it hard or soft?" During a sad moment, parents might ask children: "Where do you feel sad: in your eyes, in your throat, in your chest?" Before bedtime, parents might ask children to take deep breaths and notice how their body feels calmer after breathing slowly.

POTENTIAL BENEFITS OF MINDFULNESS TRAINING

From the perspective of the iterative reprocessing model, these age-appropriate mindfulness training activities target both top-down and bottom-up influences on self-regulation. That is, training attention to one's moment-to-moment experiences exercises top-down reflection. In addition, practice being nonjudgmental produces calmness and well-being, as does focusing on the present moment instead of, say, ruminating over a recollected source of anxiety. Thus, at both the cognitive level (attention) and the emotional level (evaluation), mindfulness training disrupts the automatic elicitation of emotional responses, resulting in greater

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